

Supplementary Material for “BrainANet: an Anatomical Brain Representation Network with Large-scale Self-supervised Learning for Brain Disease Diagnosis”

Zhenpeng Chen, Bin Jing, Benzhen Wei

S1. ADDITIONAL METHODOLOGICAL DETAILS

A. Similarity Measures for BrainANet Construction

Let $\mathbf{r}_i^{(s)} \in \mathbb{R}^d$ denote the adaptive regional representation extracted from the i th region of interest (ROI) for subject s , where d is the representation dimension. For a brain atlas containing N ROIs, the pairwise similarity between $\mathbf{r}_i^{(s)}$ and $\mathbf{r}_j^{(s)}$ was calculated to construct the subject-specific BrainANet adjacency matrix $\mathbf{A}^{(s)} \in \mathbb{R}^{N \times N}$, whose (i, j) th entry is defined as:

$$A_{ij}^{(s)} = \mathcal{S}(\mathbf{r}_i^{(s)}, \mathbf{r}_j^{(s)}), \quad (\text{S1})$$

where $\mathcal{S}(\cdot, \cdot)$ denotes the similarity measure. Four similarity measures were evaluated, including cosine similarity, Gaussian kernel similarity, Jensen–Shannon divergence-based similarity, and the Pearson correlation coefficient.

1) *Cosine Similarity*: Cosine similarity quantifies the angular agreement between two regional representations and is independent of their magnitudes. It was calculated as

$$C_{ij}^{(s)} = \frac{(\mathbf{r}_i^{(s)})^T \mathbf{r}_j^{(s)}}{\|\mathbf{r}_i^{(s)}\|_2 \|\mathbf{r}_j^{(s)}\|_2 + \epsilon}, \quad (\text{S2})$$

where $\|\cdot\|_2$ denotes the Euclidean norm and ϵ is a small constant introduced to ensure numerical stability. Because cosine similarity ranges from -1 to 1 , it was linearly rescaled to the interval $[0, 1]$:

$$A_{ij, \text{Cosine}}^{(s)} = \frac{C_{ij}^{(s)} + 1}{2}. \quad (\text{S3})$$

2) *Gaussian Kernel Similarity*: Gaussian kernel similarity measures the proximity between two representations according to their squared Euclidean distance:

$$A_{ij, \text{Gaussian}}^{(s)} = \exp\left(-\frac{\|\mathbf{r}_i^{(s)} - \mathbf{r}_j^{(s)}\|_2^2}{2\gamma^2}\right), \quad (\text{S4})$$

where $\gamma > 0$ denotes the kernel bandwidth controlling the sensitivity of the similarity value to the distance between regional representations. A smaller γ results in a faster decrease in similarity with increasing distance, whereas a larger γ produces a smoother similarity distribution.

3) *Jensen–Shannon Divergence-Based Similarity*: Because Jensen–Shannon divergence is defined for probability distributions, each regional representation was first transformed into a probability vector using the softmax function:

$$p_{i,k}^{(s)} = \frac{\exp(r_{i,k}^{(s)}/\tau)}{\sum_{\ell=1}^d \exp(r_{i,\ell}^{(s)}/\tau)}, \quad k = 1, \dots, d, \quad (\text{S5})$$

where $r_{i,k}^{(s)}$ denotes the k th element of $\mathbf{r}_i^{(s)}$, and $\tau > 0$ is a temperature parameter. Let

$$\mathbf{m}_{ij}^{(s)} = \frac{1}{2} (\mathbf{p}_i^{(s)} + \mathbf{p}_j^{(s)}). \quad (\text{S6})$$

The Jensen–Shannon divergence between the two normalized representations was calculated as

$$D_{\text{JS}}(\mathbf{p}_i^{(s)}, \mathbf{p}_j^{(s)}) = \frac{1}{2} D_{\text{KL}}(\mathbf{p}_i^{(s)} \parallel \mathbf{m}_{ij}^{(s)}) + \frac{1}{2} D_{\text{KL}}(\mathbf{p}_j^{(s)} \parallel \mathbf{m}_{ij}^{(s)}), \quad (\text{S7})$$

where

$$D_{\text{KL}}(\mathbf{p} \parallel \mathbf{q}) = \sum_{k=1}^d p_k \log\left(\frac{p_k + \epsilon}{q_k + \epsilon}\right) \quad (\text{S8})$$

denotes the Kullback–Leibler divergence. Because Jensen–Shannon divergence is a dissimilarity measure, it was converted into a similarity score as

$$A_{ij, \text{JS}}^{(s)} = 1 - \frac{D_{\text{JS}}(\mathbf{p}_i^{(s)}, \mathbf{p}_j^{(s)})}{\log 2}. \quad (\text{S9})$$

When the natural logarithm is used, the normalized similarity value lies within $[0, 1]$, with larger values indicating greater similarity.

4) *Pearson Correlation Coefficient*: The Pearson correlation coefficient measures the linear association between two regional representations after mean-centering. It was calculated as

$$P_{ij}^{(s)} = \frac{\sum_{k=1}^d (r_{i,k}^{(s)} - \bar{r}_i^{(s)}) (r_{j,k}^{(s)} - \bar{r}_j^{(s)})}{\sqrt{\sum_{k=1}^d (r_{i,k}^{(s)} - \bar{r}_i^{(s)})^2} \sqrt{\sum_{k=1}^d (r_{j,k}^{(s)} - \bar{r}_j^{(s)})^2 + \epsilon}}, \quad (\text{S10})$$

where

$$\bar{r}_i^{(s)} = \frac{1}{d} \sum_{k=1}^d r_{i,k}^{(s)} \quad (\text{S11})$$

is the mean value of the representation elements for the i th ROI. Because the Pearson correlation coefficient ranges from -1 to 1 , it was rescaled to $[0, 1]$:

$$A_{ij,\text{PCC}}^{(s)} = \frac{P_{ij}^{(s)} + 1}{2}. \quad (\text{S12})$$

For all similarity measures, the resulting adjacency matrix was symmetric, and the diagonal elements were set to one:

$$A_{ij}^{(s)} = A_{ji}^{(s)}, \quad A_{ii}^{(s)} = 0. \quad (\text{S13})$$

The resulting weighted adjacency matrix was subsequently sparsified using the predefined threshold β before being used for graph-based brain disease classification.

B. Architecture of Patch-Level CNN

Patch-Level CNN was implemented using a lightweight 3D convolutional encoder-decoder. The encoder comprised three convolutional blocks, each consisting of a 3D convolution, batch normalization, LeakyReLU activation, and max pooling, with progressively increased feature dimensions. The decoder progressively restored the spatial resolution through 3D convolution, instance normalization, LeakyReLU activation, and trilinear upsampling, followed by a $1 \times 1 \times 1$ convolution for image reconstruction. The high-level features extracted by the encoder were used as adaptive regional representations for constructing BrainANet-CNN.

C. GNN Models Implemented in This Study

To evaluate the compatibility of BrainANet with different graph-learning architectures, four representative graph neural networks were implemented, including GCN, GIN, GraphSAGE, and GlobalAttentionNet. Unless otherwise specified, each model contained two graph-convolutional layers, followed by a graph-level representation module and a multilayer classifier. A dropout rate of 0.5 was applied before the final classification layer.

1) *Architecture of GCN*: The GCN consisted of multiple graph convolutional layers, each followed by a ReLU activation. Given the node-feature matrix $\mathbf{H}^{(l)}$ at the l -th layer, the graph convolution could be expressed as:

$$\mathbf{H}^{(l+1)} = \sigma \left(\tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}} \mathbf{H}^{(l)} \mathbf{W}^{(l)} \right), \quad (\text{S14})$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ denotes the adjacency matrix with self-connections, $\tilde{\mathbf{D}}$ is its corresponding degree matrix, $\mathbf{W}^{(l)}$ is a trainable weight matrix, and $\sigma(\cdot)$ denotes the ReLU activation function. The edge weights of BrainANet were incorporated into $\tilde{\mathbf{A}}$ during message propagation.

The node representations obtained from all graph convolutional layers were concatenated:

$$\mathbf{H}_{\text{cat}} = \mathbf{H}^{(1)} \parallel \mathbf{H}^{(2)} \parallel \dots \parallel \mathbf{H}^{(L)}, \quad (\text{S15})$$

where $\parallel$ denotes feature concatenation. Because all graphs contained a fixed number of nodes, the concatenated node representations were converted into a dense matrix and flattened into a graph-level vector:

$$\mathbf{z}_G = \text{vec}(\mathbf{H}_{\text{cat}}). \quad (\text{S16})$$

The resulting vector was subsequently passed through a fully connected layer, ReLU activation, dropout, and the final classification layer.

2) *Architecture of GIN*: The GIN updated each node representation using a learnable multilayer perceptron. The node-update operation at the l -th layer was defined as

$$\mathbf{h}_i^{(l+1)} = \text{MLP}^{(l)} \left[\left(1 + \epsilon^{(l)} \right) \mathbf{h}_i^{(l)} + \sum_{j \in \mathcal{N}(i)} \mathbf{h}_j^{(l)} \right], \quad (\text{S17})$$

where $\mathbf{h}_i^{(l)}$ is the representation of node i , $\mathcal{N}(i)$ denotes its neighboring nodes, and $\epsilon^{(l)}$ is a trainable parameter.

Each MLP comprised two linear layers, ReLU activations, and batch normalization. The node representations generated by all GIN layers were concatenated and aggregated using global mean pooling:

$$\mathbf{z}_G = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \left(\mathbf{h}_i^{(1)} \parallel \mathbf{h}_i^{(2)} \parallel \dots \parallel \mathbf{h}_i^{(L)} \right), \quad (\text{S18})$$

where $\mathcal{V}$ denotes the set of graph nodes. The pooled graph representation was then passed through a fully connected classification module.

3) *Architecture of GraphSAGE*: GraphSAGE learned node representations by aggregating information from neighboring nodes. Using mean aggregation, its message-passing operation could be expressed as:

$$\mathbf{m}_i^{(l)} = \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \mathbf{h}_j^{(l)}, \quad (\text{S19})$$

followed by

$$\mathbf{h}_i^{(l+1)} = \sigma \left(\mathbf{W}_{\text{self}}^{(l)} \mathbf{h}_i^{(l)} + \mathbf{W}_{\text{neigh}}^{(l)} \mathbf{m}_i^{(l)} \right), \quad (\text{S20})$$

where $\mathbf{W}_{\text{self}}^{(l)}$ and $\mathbf{W}_{\text{neigh}}^{(l)}$ are trainable transformation matrices.

The representations generated by different GraphSAGE layers were concatenated and summarized through global mean pooling:

$$\mathbf{z}_G = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \left(\mathbf{h}_i^{(1)} \parallel \mathbf{h}_i^{(2)} \parallel \dots \parallel \mathbf{h}_i^{(L)} \right). \quad (\text{S21})$$

The resulting graph-level representation was processed by a fully connected layer, ReLU activation, dropout, and a final classification layer.

4) *Architecture of GlobalAttentionNet*: GlobalAttentionNet first employed multiple GraphSAGE layers to encode local neighborhood information. The final node representations were subsequently aggregated using an attention-based graph pooling mechanism. For node i , the normalized attention coefficient was calculated as

$$\alpha_i = \frac{\exp[g(\mathbf{h}_i)]}{\sum_{j \in \mathcal{V}} \exp[g(\mathbf{h}_j)]}, \quad (\text{S22})$$

where $g(\cdot)$ is a trainable linear gating function.

The graph-level representation was obtained as the attention-weighted sum of all node representations:

$$\mathbf{z}_G = \sum_{i \in \mathcal{V}} \alpha_i \mathbf{h}_i. \quad (\text{S23})$$

This attention-based pooling mechanism allowed the model to assign different contributions to individual brain regions. The resulting graph representation was then passed through a fully connected layer, ReLU activation, dropout, and the final classification layer.

D. Data Processing

The sMRI data were preprocessed using CAT12 toolbox (Version12.7, r1700). In the 'initial voxel-based processing' step, the sMRI images were denoised using the spatially adaptive non-local means (SANLM) filter. To improve image consistency, bias-field correction and affine registration were applied to align the denoised images with the standard space. Finally, the unified segmentation algorithm was adopted to segment the images into different tissue classes. In the 'refined voxel-based processing' step, intensity transformations were used across tissue classes to reduce the influence of regionally elevated gray matter intensities, particularly in areas such as the motor cortex, basal ganglia, and occipital lobe. Subsequently, the adaptive maximum a posteriori (AMAP) segmentation algorithm, combined with partial volume estimation, was applied to estimate the fractional contribution of each tissue class within individual voxels.

The fMRI data, selected from ABIDE and RestMDD datasets, were preprocessed using Data Processing & Analysis for Brain Imaging (DPABI, version5.5) toolbox. Specifically, the first 10 volumes were discarded, and the remaining volumes underwent slice-timing correction and realignment. Each subject's structural MRI was co-registered to the functional images, segmented, and normalized to the Montreal Neurological Institute (MNI) space. The resulting transformation parameters were applied to the functional images, which were resampled to $3 \times 3 \times 3 \text{ mm}^3$. Nuisance covariates, including six head-motion parameters, mean white-matter and cerebrospinal-fluid signals, and linear trends, were regressed out. Finally, temporal band-pass filtering at 0.01–0.10 Hz and spatial smoothing with a 4-mm full width at half maximum Gaussian kernel were performed.

E. Centered Kernel Alignment Analysis

Centered kernel alignment (CKA) was used to quantify the representational similarity between BrainANet-SwinVIT and RSFC. For each repeated experiment, graph-level latent representations were extracted from the two GAT models immediately before the final classification layer for the same set of test subjects. Let $\mathbf{X} \in \mathbb{R}^{N \times d_B}$ and $\mathbf{Y} \in \mathbb{R}^{N \times d_R}$ denote the latent representations derived from BrainANet-SwinVIT and RSFC, respectively, where N is the number of subjects and d_B and d_R are the corresponding representation dimensions. After feature centering, linear CKA was calculated as:

$$\text{CKA}(\mathbf{X}, \mathbf{Y}) = \frac{\|\mathbf{X}^T \mathbf{Y}\|_F^2}{\|\mathbf{X}^T \mathbf{X}\|_F \|\mathbf{Y}^T \mathbf{Y}\|_F}, \quad (\text{S24})$$

where $\|\cdot\|_F$ denotes the Frobenius norm. CKA values range from 0 to 1, with higher values indicating greater similarity between the representational structures of the two modalities. CKA was calculated independently for each of the five repeated experiments. Subject-level bootstrap resampling was further performed to estimate the corresponding 95% confidence intervals.

F. Definitions and Calculation of ICC(1,1) and SOG fingerprint

ICC quantified the reproducibility of extracted features by the patch-level Swin Transformer, defined as:

$$\text{ICC}(1,1) = \frac{MS_B - MS_W}{MS_B + (k-1)MS_W} \quad (\text{S25})$$

where MS_B denotes the between-subject mean square, MS_W denotes the within-subject mean square, and k is the number of repeated measurements. An ICC matrix of size $N_p * D$ was then constructed, where N_p denotes number of selected patches and D denotes the feature dimensionality.

The SOG fingerprint score was used to assess the individual identifiability of the extracted features. For a given scan pair $p = (A_p, B_p)$, the SOG score was calculated as

$$\Delta^{(p)} = \frac{1}{N_s} \sum_{i=1}^{N_s} \left[S_{ii}^{(p)} - \frac{1}{N_s - 1} \sum_{\substack{j=1 \\ j \neq i}}^{N_s} S_{ij}^{(p)} \right], \quad (\text{S26})$$

where N_s denotes the number of subjects in the test-retest dataset. The diagonal elements $S_{ii}^{(p)}$ represent within-subject similarities, whereas the off-diagonal elements $S_{ij}^{(p)}$ ($i \neq j$) represent between-subject similarities. For the p -th scan pair, the similarity was calculated as

$$S_{ij}^{(p)} = \frac{\mathbf{x}_i^{(A_p)} \cdot \mathbf{x}_j^{(B_p)}}{\|\mathbf{x}_i^{(A_p)}\|_2 \|\mathbf{x}_j^{(B_p)}\|_2}, \quad (\text{S27})$$

where $\mathbf{x}_i^{(A_p)}$ and $\mathbf{x}_j^{(B_p)}$ denote the feature vectors of subjects i and j obtained from the two scans constituting the p -th scan pair, respectively. The SOG score was calculated independently for each scan pair and then averaged across all P scan pairs:

$$\bar{\Delta} = \frac{1}{P} \sum_{p=1}^P \Delta^{(p)}, \quad (\text{S28})$$

where P denotes the total number of scan pairs. The averaged SOG score $\bar{\Delta}$ was used as the final measure of individual identifiability.

S2. SUPPLEMENTARY DEMOGRAPHIC CHARACTERISTICS

Participant demographics, including age, sex, and site distributions across the four diagnostic tasks, together with the corresponding between-group statistical comparisons, are summarized in Table SI.

S3. ADDITIONAL RESULTS

Paired performance differences and 95% confidence intervals for the network comparisons in the NC versus AD, sMCI versus pMCI, NC versus ASD, and NC versus MDD classification tasks are presented in Tables SII–SV, respectively. Additional performance and interpretability analyses for the NC versus ASD and NC versus MDD tasks are presented in Figure S1. The ROC curves and radar plots compare the classification performance of four anatomical brain networks, whereas the t-SNE visualizations illustrate the latent representations learned by GAT from BrainANet-SwinViT and iRSSN. The circular plots further present the top discriminative connections identified by two-sample t-tests and the connection-importance matrices learned by the GAT interpretability module.

S4. DATA AVAILABILITY

AOMIC-ID1000: <https://openneuro.org/datasets/ds003097>
AOMIC-PIOP1: <https://openneuro.org/datasets/ds002785>
AOMIC-PIOP2: <https://openneuro.org/datasets/ds002790>
HCP: <https://www.humanconnectome.org/study/hcp-young-adult/document/1200-subjects-data-release>
ENKI: https://fcon_1000.projects.nitrc.org/indi/pro/eNKI_RS_TRT/FrontPage.html
IXI: <https://brain-development.org/ixi-dataset/>
SLIM: https://fcon_1000.projects.nitrc.org/indi/retro/southwestuni_qiu_index.html
DLBS: https://fcon_1000.projects.nitrc.org/indi/retro/dlbs.html
MR-ART: <https://openneuro.org/datasets/ds004173>
SALD: https://fcon_1000.projects.nitrc.org/indi/retro/sald.html
CAB: <https://www.nitrc.org/projects/adultatlas>
ADNI-1: <https://adni.loni.usc.edu/>
ADNI-2: <https://adni.loni.usc.edu/>
ADNI-3: <https://adni.loni.usc.edu/>
NACC: <https://naccdata.org/>
AIBL: <https://aibl.org.au/>
REST-MDD: <https://rfmri.org/REST-meta-MDD>
ds000171: <https://github.com/OpenNeuroDatasets/ds000171>
ds003653: <https://github.com/OpenNeuroDatasets/ds003653>
ABIDE: https://fcon_1000.projects.nitrc.org/indi/abide/

TABLE SI
DEMOGRAPHIC CHARACTERISTICS ACROSS THE FOUR DISEASE-CLASSIFICATION TASKS.

Task	Group	N	Percentage	Age, mean $\pm$ SD	Age, median ([IQR])	Minimum age	Maximum age	Sex, n (%)	Age statistic	Age p value	Age effect size	Sex statistic	Sex p value	Sex Cramér's V	Site statistic	Site p value	Site Cramér's V
Dementia	CN	2715	76.9%	72.71 $\pm$ 8.39	73.00 [67.00, 78.10]	33	101	F: 1669 (61.0%); M: 1066 (39.0%)	4.978	<0.001	0.199	17.584	<0.001	0.070	204.583	<0.001	0.240
	AD	821	23.1%	74.38 $\pm$ 8.42	75.00 [69.00, 80.30]	49	95	F: 433 (52.7%); M: 388 (47.3%)									
	sMCI	469	57.1%	72.52 $\pm$ 7.91	72.80 [66.90, 78.30]	47	90.1	F: 205 (43.7%); M: 264 (56.3%)	1.906	0.057	0.134	0.691	0.406	0.029	32.469	<0.001	0.199
	pMCI	352	42.9%	73.59 $\pm$ 7.96	73.80 [69.00, 79.43]	53	96	F: 165 (46.9%); M: 187 (53.1%)									
ASD	ASD	592	53.2%	14.97 $\pm$ 9.31	11.15 [9.40, 18.00]	5.887	64	F: 181 (30.6%); M: 411 (69.4%)									
	NC	520	46.8%	14.74 $\pm$ 9.01	12.05 [9.19, 18.00]	5.128	62	F: 77 (14.8%); M: 443 (85.2%)	-0.415	0.678	-0.025	37.742	<0.001	0.184	115.187	<0.001	0.322
MDD	MDD	1270	52.9%	36.17 $\pm$ 14.54	33.00 [25.00, 45.00]	12	82	F: 806 (63.5%); M: 464 (36.5%)	-0.234	0.815	-0.010	4.783	0.029	0.045	57.954	<0.001	0.155
	NC	1130	47.1%	36.03 $\pm$ 15.40	30.35 [23.00, 47.00]	12	80	F: 667 (59.0%); M: 463 (41.0%)									

Abbreviations: AD, Alzheimer's disease; ASD, autism spectrum disorder; CN, cognitively normal; F, female; IQR, interquartile range; M, male; MDD, major depressive disorder; N, number of participants; pMCI, progressive mild cognitive impairment; sMCI, stable mild cognitive impairment; SD, standard deviation. **Notes:** Percentages were calculated within each binary classification task. Age statistics represent Welch's t values, with Hedges' g reported as the corresponding effect-size measure. Sex and site statistics represent Pearson's χ^2 values, with Cramér's V reported as the corresponding effect-size measure.

TABLE SII
PAIRED PERFORMANCE DIFFERENCES AND 95% CONFIDENCE INTERVALS FOR THE NETWORK COMPARISONS IN THE NC VERSUS AD CLASSIFICATION TASK.

Method_A	Method_B	Metric	Paired_Runs	Mean_A	Mean_B	Mean_Difference_A_minus_B	CI_Lower	CI_Upper	Mean_Difference_pp	CI_Lower_pp	CI_Upper_pp	Difference_95CI_pp
BrainANet_SwinViT	iRSSN	ACC	5	0.902	0.858	0.043	0.036	0.051	4.3395	3.6089	5.0701	4.34 [3.61, 5.07]
BrainANet_SwinViT	iRSSN	PRE	5	0.748	0.65	0.098	0.081	0.115	9.7735	8.094	11.4529	9.77 [8.09, 11.45]
BrainANet_SwinViT	iRSSN	SEN	5	0.865	0.836	0.029	0.011	0.047	2.8989	1.0986	4.6992	2.90 [1.10, 4.70]
BrainANet_SwinViT	iRSSN	SPE	5	0.913	0.865	0.048	0.035	0.06	4.7716	3.5183	6.025	4.77 [3.52, 6.03]
BrainANet_SwinViT	iRSSN	AUC	5	0.933	0.908	0.025	0.023	0.028	2.5377	2.2998	2.7756	2.54 [2.30, 2.78]
BrainANet_SwinViT	iRSSN	F1	5	0.802	0.731	0.071	0.062	0.079	7.089	6.2322	7.9457	7.09 [6.23, 7.95]
BrainANet_SwinViT	R2SN	ACC	5	0.902	0.806	0.095	0.056	0.135	9.5278	5.5747	13.4809	9.53 [5.57, 13.48]
BrainANet_SwinViT	R2SN	PRE	5	0.748	0.653	0.095	-0.075	0.265	9.5048	-7.4673	26.477	9.50 [-7.47, 26.48]
BrainANet_SwinViT	R2SN	SEN	5	0.865	0.405	0.459	0.425	0.494	45.944	42.4764	49.4116	45.94 [42.48, 49.41]
BrainANet_SwinViT	R2SN	SPE	5	0.913	0.926	-0.014	-0.075	0.047	-1.3957	-7.4728	4.6814	-1.40 [-7.47, 4.68]
BrainANet_SwinViT	R2SN	AUC	5	0.933	0.715	0.218	0.2	0.237	21.8319	19.9588	23.7049	21.83 [19.96, 23.70]
BrainANet_SwinViT	R2SN	F1	5	0.802	0.494	0.309	0.275	0.342	30.8588	27.5274	34.1901	30.86 [27.53, 34.19]
BrainANet_SwinViT	BrainANet_CNN	ACC	5	0.902	0.862	0.04	0.036	0.044	3.9741	3.5851	4.3632	3.97 [3.59, 4.36]
BrainANet_SwinViT	BrainANet_CNN	PRE	5	0.748	0.66	0.088	0.073	0.103	8.7738	7.2771	10.2705	8.77 [7.28, 10.27]
BrainANet_SwinViT	BrainANet_CNN	SEN	5	0.865	0.826	0.039	0.026	0.052	3.8977	2.5792	5.2161	3.90 [2.58, 5.22]
BrainANet_SwinViT	BrainANet_CNN	SPE	5	0.913	0.873	0.04	0.031	0.049	3.9971	3.1402	4.854	4.00 [3.14, 4.85]
BrainANet_SwinViT	BrainANet_CNN	AUC	5	0.933	0.919	0.014	0.011	0.017	1.3783	1.0906	1.6661	1.38 [1.09, 1.67]
BrainANet_SwinViT	BrainANet_CNN	F1	5	0.802	0.734	0.068	0.064	0.073	6.8271	6.3812	7.2729	6.83 [6.38, 7.27]

TABLE SIII
PAIRED PERFORMANCE DIFFERENCES AND 95% CONFIDENCE INTERVALS FOR THE NETWORK COMPARISONS IN THE sMCI VERSUS pMCI CLASSIFICATION TASK.

Method_A	Method_B	Metric	Paired_Runs	Mean_A	Mean_B	Mean_Difference_A_minus_B	CI_Lower	CI_Upper	Mean_Difference_pp	CI_Lower_pp	CI_Upper_pp	Difference_95CI_pp
BrainANet_SwinViT	iRSSN	ACC	5	0.71	0.486	0.224	0.212	0.237	22.4356	21.1823	23.6889	22.44 [21.18, 23.69]
BrainANet_SwinViT	iRSSN	PRE	5	0.626	0.445	0.181	0.164	0.197	18.0662	16.4018	19.7306	18.07 [16.40, 19.73]
BrainANet_SwinViT	iRSSN	SEN	5	0.777	0.896	-0.118	-0.149	-0.088	-11.8333	-14.8559	-8.8107	-11.83 [-14.86, -8.81]
BrainANet_SwinViT	iRSSN	SPE	5	0.661	0.187	0.474	0.439	0.509	47.4089	43.8972	50.9206	47.41 [43.90, 50.92]
BrainANet_SwinViT	iRSSN	AUC	5	0.792	0.662	0.13	0.118	0.142	13.0351	11.8314	14.2389	13.04 [11.83, 14.24]
BrainANet_SwinViT	iRSSN	F1	5	0.693	0.595	0.099	0.088	0.109	9.8508	8.7557	10.9458	9.85 [8.76, 10.95]
BrainANet_SwinViT	R2SN	ACC	5	0.71	0.513	0.197	0.171	0.224	19.7424	17.0504	22.4343	19.74 [17.05, 22.43]
BrainANet_SwinViT	R2SN	PRE	5	0.626	0.391	0.235	0.159	0.31	23.4577	15.9053	31.0102	23.46 [15.91, 31.01]
BrainANet_SwinViT	R2SN	SEN	5	0.777	0.381	0.397	0.1	0.694	39.6667	9.9732	69.3601	39.67 [9.97, 69.36]
BrainANet_SwinViT	R2SN	SPE	5	0.661	0.609	0.052	-0.124	0.228	5.2227	-12.3694	22.8148	5.22 [-12.37, 22.81]
BrainANet_SwinViT	R2SN	AUC	5	0.792	0.545	0.247	0.179	0.315	24.6686	17.8686	31.4686	24.67 [17.87, 31.47]
BrainANet_SwinViT	R2SN	F1	5	0.693	0.372	0.321	0.136	0.506	32.1217	13.6227	50.6206	32.12 [13.62, 50.62]
BrainANet_SwinViT	BrainANet_CNN	ACC	5	0.71	0.678	0.032	0.02	0.045	3.2319	1.9956	4.4681	3.23 [2.00, 4.47]
BrainANet_SwinViT	BrainANet_CNN	PRE	5	0.626	0.605	0.021	0.006	0.037	2.1425	0.5558	3.7293	2.14 [0.56, 3.73]
BrainANet_SwinViT	BrainANet_CNN	SEN	5	0.777	0.681	0.096	0.083	0.109	9.6111	8.2842	10.938	9.61 [8.28, 10.94]
BrainANet_SwinViT	BrainANet_CNN	SPE	5	0.661	0.675	-0.014	-0.039	0.01	-1.417	-3.8539	1.0199	-1.42 [-3.85, 1.02]
BrainANet_SwinViT	BrainANet_CNN	AUC	5	0.792	0.746	0.046	0.04	0.052	4.5597	3.9513	5.1682	4.56 [3.95, 5.17]
BrainANet_SwinViT	BrainANet_CNN	F1	5	0.693	0.641	0.053	0.043	0.063	5.2781	4.2985	6.2576	5.28 [4.30, 6.26]

TABLE SIV
PAIRED PERFORMANCE DIFFERENCES AND 95% CONFIDENCE INTERVALS FOR THE NETWORK COMPARISONS IN THE NC VERSUS ASD CLASSIFICATION TASK.

Method_A	Method_B	Metric	Paired_Runs	Mean_A	Mean_B	Mean_Difference_A_minus_B	CI_Lower	CI_Upper	Mean_Difference_pp	CI_Lower_pp	CI_Upper_pp	Difference_95CI_pp
BrainANet_SwinViT	iRSSN	ACC	5	0.574	0.566	0.008	-0.026	0.042	0.7882	-2.6327	4.2091	0.79 [-2.63, 4.21]
BrainANet_SwinViT	iRSSN	PRE	5	0.645	0.614	0.031	0.011	0.052	3.1429	1.0591	5.2267	3.14 [1.06, 5.23]
BrainANet_SwinViT	iRSSN	SEN	5	0.499	0.567	-0.069	-0.165	0.028	-6.8582	-16.5243	2.8079	-6.86 [-16.52, 2.81]
BrainANet_SwinViT	iRSSN	SPE	5	0.666	0.565	0.101	0.044	0.158	10.0873	4.3832	15.7915	10.09 [4.38, 15.79]
BrainANet_SwinViT	iRSSN	AUC	5	0.618	0.554	0.063	0.049	0.078	6.3428	4.9145	7.7711	6.34 [4.91, 7.77]
BrainANet_SwinViT	iRSSN	F1	5	0.558	0.589	-0.031	-0.098	0.037	-3.0983	-9.847	3.6505	-3.10 [-9.85, 3.65]
BrainANet_SwinViT	R2SN	ACC	5	0.574	0.449	0.125	0.091	0.159	12.4926	9.1086	15.8766	12.49 [9.11, 15.88]
BrainANet_SwinViT	R2SN	PRE	5	0.645	0.494	0.151	0.129	0.173	15.1154	12.939	17.2918	15.12 [12.94, 17.29]
BrainANet_SwinViT	R2SN	SEN	5	0.499	0.142	0.357	0.234	0.479	35.6553	23.4499	47.8606	35.66 [23.45, 47.86]
BrainANet_SwinViT	R2SN	SPE	5	0.666	0.823	-0.157	-0.26	-0.054	-15.6769	-25.9796	-5.3741	-15.68 [-25.98, -5.37]
BrainANet_SwinViT	R2SN	AUC	5	0.618	0.575	0.043	0.023	0.063	4.3118	2.3012	6.3225	4.31 [2.30, 6.32]
BrainANet_SwinViT	R2SN	F1	5	0.558	0.22	0.338	0.26	0.417	33.8208	25.9685	41.6731	33.82 [25.97, 41.67]
BrainANet_SwinViT	BrainANet_CNN	ACC	5	0.574	0.539	0.035	0.001	0.069	3.4877	0.0607	6.9147	3.49 [0.06, 6.91]
BrainANet_SwinViT	BrainANet_CNN	PRE	5	0.645	0.609	0.036	0.01	0.061	3.5837	1.042	6.1254	3.58 [1.04, 6.13]
BrainANet_SwinViT	BrainANet_CNN	SEN	5	0.499	0.447	0.051	-0.05	0.153	5.1346	-5.0096	15.2789	5.13 [-5.01, 15.28]
BrainANet_SwinViT	BrainANet_CNN	SPE	5	0.666	0.651	0.015	-0.056	0.085	1.4847	-5.5599	8.5293	1.48 [-5.56, 8.53]
BrainANet_SwinViT	BrainANet_CNN	AUC	5	0.618	0.606	0.012	-0.007	0.03	1.1565	-0.7085	3.0214	1.16 [-0.71, 3.02]
BrainANet_SwinViT	BrainANet_CNN	F1	5	0.558	0.516	0.043	-0.024	0.11	4.2745	-2.4442	10.9932	4.27 [-2.44, 10.99]

TABLE SV
 PAIRED PERFORMANCE DIFFERENCES AND 95% CONFIDENCE INTERVALS FOR THE NETWORK COMPARISONS IN THE NC VERSUS MDD
 CLASSIFICATION TASK.

Method_A	Method_B	Metric	Paired_Runs	Mean_A	Mean_B	Mean_Difference_A_minus_B	CI_Lower	CI_Upper	Mean_Difference_pp	CI_Lower_pp	CI_Upper_pp	Difference_95CI_pp
BrainANet_SwinVIT	iRSSN	ACC	5	0.578	0.488	0.089	0.077	0.101	8.9296	7.7115	10.1477	8.93 [7.71, 10.15]
BrainANet_SwinVIT	iRSSN	PRE	5	0.588	0.525	0.062	0.045	0.08	6.2497	4.4574	8.0421	6.25 [4.46, 8.04]
BrainANet_SwinVIT	iRSSN	SEN	5	0.68	0.346	0.335	0.268	0.402	33.4803	26.7885	40.1721	33.48 [26.79, 40.17]
BrainANet_SwinVIT	iRSSN	SPE	5	0.462	0.648	-0.186	-0.272	-0.101	-18.6384	-27.1604	-10.1163	-18.64 [-27.16, -10.12]
BrainANet_SwinVIT	iRSSN	AUC	5	0.59	0.5	0.09	0.063	0.116	8.9648	6.282	11.6475	8.96 [6.28, 11.65]
BrainANet_SwinVIT	iRSSN	F1	5	0.63	0.417	0.213	0.187	0.239	21.3041	18.6826	23.9256	21.30 [18.68, 23.93]
BrainANet_SwinVIT	R2SN	ACC	5	0.578	0.511	0.067	0.05	0.085	6.7055	4.9602	8.4509	6.71 [4.96, 8.45]
BrainANet_SwinVIT	R2SN	PRE	5	0.588	0.558	0.03	0.008	0.052	2.995	0.7543	5.2357	3.00 [0.75, 5.24]
BrainANet_SwinVIT	R2SN	SEN	5	0.68	0.378	0.302	0.168	0.436	30.2205	16.8175	43.6234	30.22 [16.82, 43.62]
BrainANet_SwinVIT	R2SN	SPE	5	0.462	0.659	-0.197	-0.345	-0.049	-19.6994	-34.508	-4.8908	-19.70 [-34.51, -4.89]
BrainANet_SwinVIT	R2SN	AUC	5	0.59	0.521	0.069	0.041	0.097	6.8987	4.0601	9.7373	6.90 [4.06, 9.74]
BrainANet_SwinVIT	R2SN	F1	5	0.63	0.443	0.187	0.103	0.27	18.6557	10.3443	26.9671	18.66 [10.34, 26.97]
BrainANet_SwinVIT	BrainANet_CNN	ACC	5	0.578	0.524	0.054	0.035	0.072	5.3561	3.5122	7.2	5.36 [3.51, 7.20]
BrainANet_SwinVIT	BrainANet_CNN	PRE	5	0.588	0.549	0.038	0.025	0.052	3.838	2.4765	5.1996	3.84 [2.48, 5.20]
BrainANet_SwinVIT	BrainANet_CNN	SEN	5	0.68	0.564	0.117	0.038	0.195	11.6535	3.8339	19.4732	11.65 [3.83, 19.47]
BrainANet_SwinVIT	BrainANet_CNN	SPE	5	0.462	0.479	-0.017	-0.082	0.047	-1.7153	-8.1742	4.7436	-1.72 [-8.17, 4.74]
BrainANet_SwinVIT	BrainANet_CNN	AUC	5	0.59	0.547	0.043	0.017	0.07	4.3272	1.6976	6.9568	4.33 [1.70, 6.96]
BrainANet_SwinVIT	BrainANet_CNN	F1	5	0.63	0.555	0.075	0.035	0.115	7.5144	3.4924	11.5363	7.51 [3.49, 11.54]

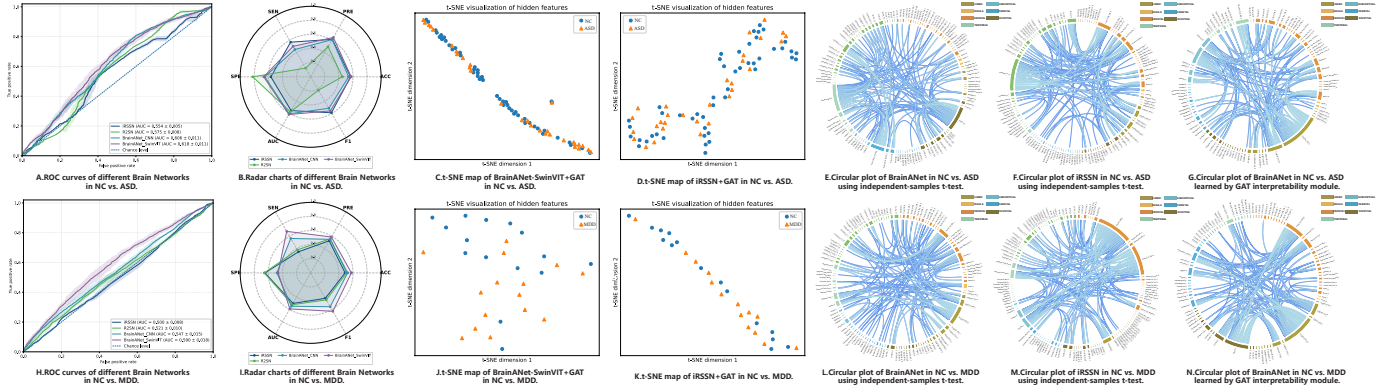


Fig. S1. ROC curves and radar plots of four anatomical brain networks evaluated using the GAT model for NC versus ASD (A, B) and NC versus MDD (H, I). t-SNE visualizations of latent representations learned by GAT from BrainANet-SwinVIT and iRSSN for NC versus ASD (C, D) and NC versus MDD (J, K). Circular plots of the top 100 discriminative connections identified using two-sample t-tests from BrainANet-SwinVIT and iRSSN for NC versus ASD (E, F) and NC versus MDD (L, M), respectively. Circular plots of the top 100 connections identified from the connection-importance matrices learned by the GAT interpretability module for BrainANet-SwinVIT in NC versus ASD (G) and NC versus MDD (N).